



MINING DEALER BEST PRACTICE

Installation of New Rod Pin Bearing by Cooling

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

In Caterpillar Service Information System, there is a procedure for an installation of a new rod pin bearing. In this procedure SEBF8242, the pin bearing of the rod is heated between 177 to 260 °C by using 6V-6996 Induction Heater.

Borusan Makina Kazakhstan installs the pin bearing of the rod by cooling bearing in a freezer instead of heating the rod. It significantly reduces the time spent for the installation.

This method of installation of new rod pin bearing is applicable for any connecting rods, the general procedure is the same as in SEBF8242, but the bearing of the rod is cooled instead of being heated.

2.0 Best Practice Description

In order to install the new rod pin bearing, all the steps (except heating) included in SEBF8242 reuse and salvage guideline are performed. The current method is to take the rods which are at the room temperature (they are at roughly 26.6°C). After that, the bearings are cooled to approximately -70°C in a freezer and then they are assembled, by using 5P-8639 Hydraulic Press.

In our company we are successfully performing this process and for today, no problems occurred.

3.0 Implementation Steps

Below is an example of installing of rod pin bearing by cooling for the connecting rod of 3500 series engine. Refer to SEBF8242, for detailed procedures.

- 1) Make a visual inspection of the rod.



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- 2) Remove the old pin bearing by using 5P-8639 Hydraulic Press.



- 3) Clean the inner diameter of the rod.



- 4) Cool the new pin bearing of the rod in the freezer, approximately to -70°C – (-75°C)



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5) Install the new bearing



6) Perform Retention test by using 5P-8639 Hydraulic Press.



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4.0 Benefits

- The heating of connecting rod requires time to heat, and every rod is heated separately. While in cooling, all the bearings can be cooled at the same time in the freezer and it significantly saves time.
- Cooling is also safer than heating. When heating the rod between 177 to 260 °C the operator must be very careful, because overheating may damage the rod. Also heating might harm the operator.

5.0 Resources Required

- The required tools are the same as in SEBF8242, but there is a freezer instead of heater.

Part Number	Part Description
5P-8639	Hydraulic Press
5P-8643	Post Assembly
5P-8642	Clamp Pin
5P-9725	Hydraulic Cylinder Gp
5P-8655	Tool Group for Standard Connecting Rods
5P-0159	Cap (part of 5P-9725)
1P-2375	Quick Connect Coupler ⁽¹⁾
1P-7437	Protection Cap ⁽¹⁾
222-3074	Die Grinder
4C-8629	Mandrel
8T-0820	Hydraulic Pressure Gauge
6V-8237	Washer
N/a	Freezer

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

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8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to:

Almaz KENZHETAYEV
Machine Shop Supervisor
Borusan Makina Kazakhstan LLP
akenzhetaev@borusan.com

Yevgeniy MARSHAL
Machine Shop Team Leader
Borusan Makina Kazakhstan LLP
Ymarshal@borusan.com

Valeriy DROVORUB
Machine Shop Fitter
Borusan Makina Kazakhstan LLP

Vitaly VARYANITSA
Machine Shop Turner
Borusan Makina Kazakhstan LLP

This Best Practice was written by:

Nurgissa DUISEMBIYEV
CRC Quality Engineer
Borusan Makina Kazakhstan LLP
nduisembiyev@borusan.com
+7 (707) 123 58 42

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